



**Course - DSE & DBD**  
**course code - 25SC1302E**

# SignEase

Breaking barriers through sign language — a full-stack translator with real-time, in-browser hand recognition.

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## THE PROBLEM

# Communication shouldn't require a shared language.



### **Schools & Colleges**

Deaf and hard-of-hearing students often lack real-time tools to communicate in mixed classrooms.



### **Workplaces**

Everyday professional interactions default to spoken/written language, excluding sign-language users.



### **Public Spaces**

Hospitals, transit, and government offices rarely offer any live translation aid at all.

## WHAT SIGNEASE DOES

# Everything you need to communicate



### Real-Time Translation

Live camera feed with hand tracking.



### Sign Dictionary

Searchable library of common signs.



### User Accounts

Profiles, favorites, and feedback.

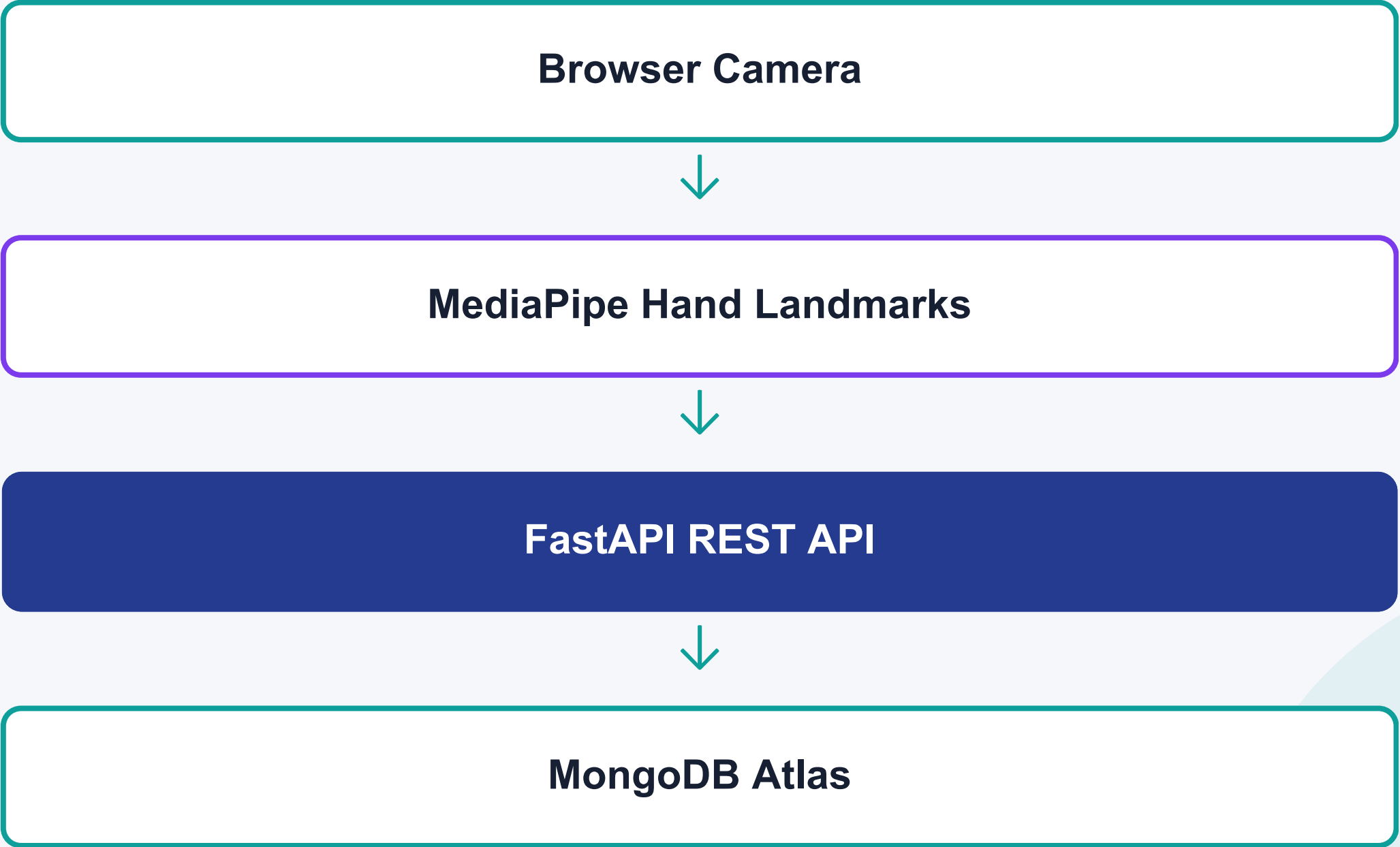


### Learning Mode

Practice signs and track progress.

# Tech stack & architecture

|                |                                              |
|----------------|----------------------------------------------|
| Frontend       | HTML / CSS / JavaScript (SPA)                |
| Backend        | FastAPI (Python), JWT auth                   |
| Database       | MongoDB (Atlas / Compass)                    |
| ML Recognition | MediaPipe Hands + k-NN, in-browser           |
| Hosting        | Render (backend) + static hosting (frontend) |



# From hand pose to predicted sign

1

## Capture landmarks

MediaPipe Hands (pretrained) extracts 21 (x, y, z) points per hand, live in the browser.

2

## Collect labeled samples

A built-in Train Signs tool records landmark snapshots labeled with a sign name.

3

## Normalize

Each sample is translated to the wrist origin and scaled, making it position/size invariant.

4

## Classify (k-NN)

Live landmarks are compared to saved samples; nearest neighbors vote on the predicted sign.

## OVERVIEW

# Six course outcomes, six implemented topics

| CO  | Topic chosen                        | Where it lives in SignEase                                   |
|-----|-------------------------------------|--------------------------------------------------------------|
| C01 | Index Design                        | Automatic MongoDB indexes created on startup                 |
| C02 | MongoDB Document Engineering (CRUD) | translations / favorites / users collections via Motor       |
| C03 | Authentication & Security (JWT)     | /auth/login issuing bearer tokens, bcrypt password hashing   |
| C04 | Architectural Thinking (applied)    | Domain-separated routes/services inside one FastAPI app      |
| C05 | API Gateway pattern (applied)       | FastAPI as single entry point: routing, auth, error handling |
| C06 | Documentation & Final Showcase      | README, render.yaml/Procfile, this presentation              |

# Future work

- 1 Collect 30-50+ samples per sign to improve classifier accuracy
- 2 Support motion-based signs using landmark sequences, not single frames
- 3 Two-hand recognition for signs that use both hands
- 4 Deploy backend to Render with MongoDB Atlas for public access